

Locating a CompTIA+ Study Resource in a Personal Obsidian Vault:

An Exploratory Retrieval and Answer-Key Integrity Audit

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Personal Knowledge Management Study — Session Record

Session date recorded in vault log

Abstract

This paper reports a small-scale, single-participant audit carried out on one study note held in a personal Obsidian vault. The work proceeded in four stages. First, a case-insensitive recursive retrieval was executed against the vault to locate CompTIA+ study material; one matching title was returned — a Security+ textbook — present both as a source PDF and as a directory of fifteen extracted per-module notes. Second, the note for Module 7, “Identity and Access Management” (SY0-701), was selected by the participant, read in full (471 lines), and audited for internal coherence; two defects were surfaced, an internal contradiction between a statement of industry password-expiry practice and an explicit recommendation against expiry, and a malformed review question offering two textually identical answer options. Third, twenty-five examination items drawn from the module were answered under a drill protocol in which every answer was scored and supported by a line-level citation into the source note; all twenty-five items were resolved against the note, and one further defect was identified, an item demanding two selections for which only one was supportable. Fourth, a purported answer key supplied by the participant was subjected to integrity analysis, which found the key to be corrupt: across roughly three quarters of its entries the printed option letter contradicted the accompanying rationale, and four of its twenty-five entries were topically unrelated to any supplied item. Re-mapping the key by subject matter rather than by position yielded coverage of twenty-one of the twenty-five items. Scoring the recorded responses against the reconstructed intended answers rather than the printed letters produced sixteen confirmed correct, one confirmed incorrect, three disputed, and four unverifiable responses. The methodological conclusion is that a printed option letter is not self-validating, that any accompanying rationale must be checked for consistency against both the item and the proposed answer before use, and that subject-matter re-mapping is necessary whenever positional alignment between item and key has been broken. The disputed and unverifiable items are retained as open findings rather than resolved silently.

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1 Introduction

1.1 Motivation

Personal knowledge management systems increasingly serve as the substrate for exam preparation. When a certification guide is imported into a vault and decomposed into per-module notes, two properties become immediately consequential. The first is *structural*: the learner must be able to find the intended artifact, and must be able to identify which fragment of a large imported corpus corresponds to the objective currently under study. The second is *epistemic*: material presented in the form of an assessment, together with any key supplied with it, must be internally consistent, otherwise the learner rehearses against error rather than against the underlying subject.

The two properties fail in different ways and for different reasons. Retrieval failure is usually benign — a naming or indexing problem that produces an empty or misleading result set. Assessment-integrity failure is subtler and more damaging, because a defective item or a defective key is indistinguishable from a valid one at the moment of use. A learner who drills against a corrupt key can leave a study session with a systematically distorted model of the material, and can do so while experiencing the session as successful.

1.2 Research questions

This session record is organised around three questions:

1. **Retrieval.** Can the intended certification study material be located unambiguously in a personal vault, and does the retrieved material decompose into an addressable per-module structure?

2. **Item quality.** What defects, if any, are present in the audited module note and in the examination items derived from it, as detected by direct reading and by line-level citation into the source?
3. **Key integrity.** When a purported answer key is cross-referenced against the items it purports to answer, what is its condition, and what scoring follows if answers are graded against reconstructed intent rather than against printed option letters?

The third question is the load-bearing one. The analysis reported in Section 3.3 is descriptive of a single audit; it makes no claim about answer keys in general, and no external literature is cited in support of any observation made here.

2 Methods

2.1 Setting and materials

The audit was performed on a single participant’s personal Obsidian vault. The materials are those retrieved within that vault; no external dataset, corpus, or public benchmark was involved. Counts reported below (fifteen module notes; 471 lines; twenty-five items; twenty-five key entries) are counts of the objects actually handled in the session.

2.2 Retrieval procedure

A case-insensitive recursive search was executed across the vault for the certification study material. The result set was recorded before any content was read, so that the reported structure reflects retrieval rather than expectation.

2.3 Module selection and reading

The participant narrowed scope to Module 7, “Identity and Access Management” (SY0-701). The corresponding note was read in full, 471 lines, and its structure was recorded as being organised around three objectives:

1. authentication credential types;
2. authentication best practices; and
3. access control schemes.

Full reading rather than sampling was performed because the audit questions concern internal consistency across the whole note; a contradiction of the kind reported in Section 3.2 is invisible to a partial read.

2.4 Citation protocol for item resolution

Twenty-five examination items were drawn from the module. The protocol adopted for answering them was as follows.

1. Each item is answered against the source note alone.
2. Every answer is scored — marked correct or incorrect — and is accompanied by a citation giving the specific line or lines in the source note that license the answer.

3. An answer is recorded as resolved only if a supporting span can be identified; otherwise the item is flagged as unresolved.

The protocol is deliberately conservative. Requiring a locatable span converts a judgement of plausibility into a judgement of licence, and it makes the disagreement cases in Section 3.3 visible rather than tacit.

2.5 Key integrity analysis

A purported answer key was subsequently supplied for cross-referencing. It was subjected to three successive analyses.

1. **Internal consistency check.** For each entry, the printed option letter was compared against the accompanying explanatory rationale. An entry is *self-contradictory* when the letter selects one option while the rationale endorses a different one.
2. **Topical correspondence check.** Each key entry was compared against the supplied items for subject-matter correspondence, independent of position. An entry is *off-topic* when no supplied item shares its subject matter.
3. **Subject-matter re-mapping.** Where positional alignment between item and key is unreliable, the key was re-indexed by subject matter to establish which items are addressed at all, and hence which are left unaddressed.

2.6 Grading rule

Where the key's own rationale supported the recorded response while its printed letter did not, the rationale was treated as authoritative and the response was upheld. This rule is stated in advance because it determines the reported tally; it is applied consistently, and the items it does not settle are reported as disputed rather than being forced into a verdict.

3 Results

3.1 Retrieval outcomes

The recursive case-insensitive search returned exactly one matching title: a Security+ textbook by Mark Ciampa. The material was present in two forms, namely a source PDF, and a directory of fifteen extracted per-module notes. Retrieval was therefore unambiguous, and the imported material had decomposed into a per-module structure that made a single objective addressable by name.

3.2 Defects in the note and in the items

The initial audit of the full module note surfaced two defects.

D1: internal contradiction on password expiry. The note contains a statement of prevailing industry practice on password expiration that is contradicted by an explicit recommendation against expiry made elsewhere in the same note. The two statements cannot both be operative guidance for a learner.

D2: malformed review question. One review question presents two textually identical answer options. Such an item has no determinate answer as written, and cannot be scored.

A third defect, *D3*, was identified during the item-resolution pass rather than during the initial read:

D3: unsupported multi-select item. One examination item demands two selections, but only one of the required selections is supportable from the source. The item is therefore unanswerable at its specified arity.

All twenty-five items were resolved against the note under the citation protocol. *Caveat: the summary of record does not preserve item-level identifiers, the ordering of items, or the verbatim text of the affected items; defects D1–D3 are reported here descriptively at the level of defect type.* The structural finding is that the module note supports item construction but does not guarantee item well-formedness: contradiction, degenerate option sets, and arity mismatches all survive a full read of the source.

3.3 Answer-key integrity

Integrity analysis of the purported key produced the following condition profile.

Property	Observed condition
Entries	25
Entries whose printed letter contradicts the accompanying rationale	roughly three quarters of entries
Entries topically unrelated to any supplied item	4
Items addressed by the key, under subject-matter re-mapping	21 of 25
Items left unaddressed by the key	4

The key is therefore corrupt on two independent axes. It is *internally* inconsistent at the level of most of its entries, and *externally* misaligned in that a fifth of the key’s volume is spent on material that appears nowhere in the item set. Positional alignment between the key and the item set is not merely noisy; it cannot be trusted as an indexing assumption, which is why re-mapping by subject matter was required.

3.4 Scoring against reconstructed intent

Grading the recorded responses against the reconstructed intended answers, rather than against the printed letters, yields the following tally over the twenty-five items.

Outcome	Items
Confirmed correct	16
Confirmed incorrect	1
Disputed	3
Unverifiable	4
Total	25

The single confirmed error. The one conceded error involved rejecting an option describing a comparison against a precomputed hash database, on the grounds that the source describes plaintext word candidates. The rejection was an over-fit to the source: the source does not mention rainbow tables, and the correct option was the precomputed-database one. This case illustrates the failure mode of citation discipline when it is applied as a literalness test rather than as a licence test — a span that fails to mention a competing concept is not a span that licenses its exclusion.

The three disputed items. These concern (i) the relative security ranking of time-based versus event-based one-time passwords, (ii) the inclusion of password expiration as a correct practice against the source’s explicit recommendation against it, and (iii) the specialisation of biometric scanners. Each is preserved as open. Item (ii) is directly entangled with defect D1: if the note contradicts itself on expiry, an item touching expiry cannot be settled from that note without first resolving the contradiction.

The four unverifiable items. These are the items the key does not address. Unverifiable here means that no trustworthy adjudication exists, not that the recorded responses were wrong.

3.5 Tally reconciliation

Table 1 restates the outcome distribution and the corresponding proportion, so that the relationship between the resolved and unresolved portions of the item set is explicit.

Outcome	Items	Share of 25
Confirmed correct	16	0.64
Confirmed incorrect	1	0.04
Disputed	3	0.12
Unverifiable	4	0.16
Total	25	1.00

Table 1: Outcome distribution of the twenty-five items under grading against reconstructed intended answers. Shares are computed from the counts; no external or published rate is invoked.

Seven of the twenty-five items — the three disputed and the four unverifiable — are therefore not settled by this audit. That fraction is itself the finding: a key that is internally inconsistent on most entries cannot adjudicate the items it does not reach, and the two failure modes compound rather than substitute for one another.

4 Discussion

4.1 On the precedence of rationale over letter

The clearest methodological outcome of the audit is the ordering it forces. A printed option letter is an index into a set of options; it carries no semantics of its own and is trivially vulnerable to transposition, to version drift between an item and its key, and to mechanical regeneration. An accompanying rationale, by contrast, states a reason, and a reason can be checked: against the item for relevance, against the source for licence, and against the letter for consistency. In roughly three quarters of the entries examined here, the rationale was the sound artefact and the letter was not, and the responses in those entries were upheld on the strength of the rationale.

This is a narrow empirical observation about one key, not a general claim. Its value is as a worked demonstration of a check that is cheap to perform and expensive to omit.

4.2 On re-mapping when positional alignment fails

Positional alignment between an item set and a key is an assumption, not an observation. Here it was false, both because the key is internally inconsistent and because four of its entries address material absent from the item set. Once positional indexing is abandoned, the key must be re-indexed by subject matter, and that re-indexing is what makes the four unaddressed items visible. Under positional grading those items would have been scored against whatever text happened to occupy their slot; under subject-matter re-mapping they are correctly recognised as unadjudicated.

4.3 On the asymmetry of defects

The three note-and-item defects D1–D3 and the key defects are of different kinds and resist a single remedy. D1 is a contradiction internal to the source, so no amount of scrutiny of the assessment layer repairs it; the source must first be reconciled. D2 and D3 are construction faults — a degenerate option set and an arity mismatch — and are properties of the item rather than of the note. The key defects are properties of a third artefact entirely. The practical implication is that “is this study material correct?” is not one question but at least three, over three artefacts that can fail independently.

4.4 Limitations

The audit is a single-participant, single-module, single-session exercise over twenty-five items drawn from one note. Nothing here is a sample from a population, and the proportions in Table 1 describe this item set only. No independent second annotator was used, so no inter-rater agreement figure can be reported. The item text, item identifiers, and the verbatim text of the defective items were not preserved in the session record, which is why D1–D3 are stated by defect type rather than quoted. No external literature is cited, and no empirical claim about answer keys in general is advanced or supported.

4.5 Preserved open findings

Consistent with the stated constraints, the following are recorded as open and are not resolved here:

- the relative security ranking of time-based versus event-based one-time passwords;
- the status of password expiration, which is entangled with the unresolved contradiction D1;
- the specialisation of biometric scanners; and
- the four items that the key does not address, for which no adjudication exists.

5 Conclusion

A case-insensitive recursive retrieval located the intended CompTIA+ material unambiguously in a personal Obsidian vault, finding a Security+ textbook present both as a source PDF and

as fifteen extracted per-module notes, and made a single module addressable by name. A full read of the 471-line note for Module 7, “Identity and Access Management” (SY0-701), together with citation-backed resolution of twenty-five derived items, surfaced three defects: an internal contradiction on password-expiry practice, a review question with two identical options, and a multi-select item of unsupportable arity.

The central finding concerns the purported answer key. It is corrupt on two independent axes: the printed option letter contradicts the accompanying rationale in roughly three quarters of its entries, and four of its twenty-five entries address no supplied item. Re-mapping by subject matter rather than by position established coverage of twenty-one of the twenty-five items. Grading against reconstructed intent rather than printed letters yields sixteen confirmed correct, one confirmed incorrect, three disputed, and four unverifiable outcomes, the single error being an over-fit to the source that excluded a precomputed-hash-database comparison the source never mentioned.

The transferable lesson is procedural rather than substantive. A printed option letter is not self-validating. A rationale must be checked for consistency with both the item and the proposed answer before it is trusted, and where it is consistent while the letter is not, the rationale governs. And whenever positional alignment between an item set and a key is broken, subject-matter re-mapping is not an optional refinement but the precondition for knowing which items have been adjudicated at all.

References

- [1] Author-supplied study note, “Module 7: Identity and Access Management (SY0-701),” personal Obsidian vault; 471 lines; read in full during the audited session.
- [2] Author-supplied directory of fifteen extracted per-module notes derived from a Security+ textbook by Mark Ciampa, held in the author’s personal Obsidian vault alongside the source PDF.